CLAIMS

What is claimed is:

1. An active data transmission system, comprising:

    - a connector housing;
    - a two-dimensional array of individually addressable electromagnetic radiation emitters contained within the connector housing;
    - a high-performance processing unit, such as a GPU, FPGA, or ASIC, contained within the same connector housing;
    - an encoder circuit configured to receive processed digital data from the integrated processing unit and to generate a sequence of frames by mapping said processed data to the emission characteristics of the emitters in the array; and
    - wherein the integrated processing unit is configured to receive an initial stream of digital data from a host computer and to perform a computational task on said data in-flight before it is passed to the encoder circuit.

2. The system of claim 1, further comprising a receiver unit, the receiver unit including:

    - a second connector housing;
    - an array of electromagnetic radiation sensors contained within the second connector housing, configured to capture the sequence of frames; and
    - a decoder circuit configured to generate a replica of the processed digital data by interpreting the emission characteristics of the captured frames.

3. The system of claim 2, wherein the receiver unit further comprises a second high-performance processing unit contained within the second connector housing, configured to perform a computational task on the data after it is captured by the sensor array.

4. The system of claim 1, wherein the computational task performed by the integrated processing unit is selected from the group consisting of: data filtering, data transformation, data compression, data decompression, data encoding, data decoding, and network routing.

5. The system of claim 1, wherein the data transmission rate of the system exceeds the memory bandwidth of the host computer.

6. The system of claim 1, wherein the integrated processing unit is configured to perform a pixel-based network routing task, wherein said task comprises assigning data packets to specific emission characteristics based on their network destination.

7. The system of claim 6, wherein the emission characteristic used for routing is the wavelength or color of the emitted radiation.

8. A method for active data transmission, comprising the steps of:

    - receiving, at a connector housing of a data cable, a stream of digital data from a host computer;
    - processing, by a high-performance processing unit located inside the connector housing, said stream of digital data to generate a processed data stream;
    - encoding, within the same connector housing, the processed data stream into a sequence of frames, wherein each frame comprises a plurality of emission channels corresponding to a two-dimensional array of emitters; and
    - emitting said sequence of frames from the two-dimensional array of emitters.

9. The method of claim 8, wherein the step of processing the data comprises executing a pixel-based network routing algorithm to determine the emission characteristics for each channel based on a network destination.

10. The method of claim 8, wherein the data is transmitted at a rate that is higher than the rate at which the host computer can load data from its main memory.